Matrix Selection and Indexing

Just like with vectors, we use the square bracket notation to select elements from a matrix. Since we have two dimensions to work with, we'll use a comma to separate our indexing for each dimension.

So the syntax is then:

example.matrix[*rows*,*columns*]

where the index notation (e.g. 1:5) is put in place of the *rows* or *columns* . If either *rows* or *columns* is left blank, then we are selecting all the rows and columns.

Let's work through some examples:

In [1]:

```
mat <- matrix(1:50,byrow=TRUE,nrow=5)
```

In [2]:

```
mat
```

Out[2]:

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

In [3]:

```
# Grab first row
mat[1,]
```

Out[3]:

1 2 3 4 5 6 7 8 9 10

In [4]:

```
#Grab first column
mat[,1]
```

Out[4]:

1 11 21 31 41

In [6]:

```
# Grab first 3 rows
mat[1:3,]
```

Out[6]:

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

21	22	23	24	25	26	27	28	29	30

In [8]:

```
# Grab top left rectangle of:
# 1,2,3
# 11,12,13
#
mat[1:2,1:3]
```

Out[8]:

1	2	3
11	12	13

In [9]:

```
# Grab last two columns
mat[,9:10]
```

Out[9]:

9	10
19	20
29	30
39	40
49	50

In [14]:

```
# Grab a center square of:
# 15,16
# 25,26
mat[2:3,5:6]
```

Out[14]:

15	16
25	26

Great job! You should practice this! Just pick out blocks from the matrix and see if you can successfully index them. Up next we will learn about Matrix Arithmetic!